Homework 1

Stat 155 Fall 2025

due Friday September 4 2025 11:59pm on gradescope

Instructions: You can earn extra credit by typing up your solutions in LaTeX. I suggest you use Overleaf as a LaTeX editor. Please see the **Academic Integrity** section of the syllabus about rules for collaborating or using outside references on your homework. Submit your homework to gradescope as a pdf. Late homework will not be accepted.

- Problem 1.1
- Exercise 2.3
- Exercise 2.5
- Exercise 2.8
- Problem 2.1 (note in part c (H) indicates that the book provides a hint)

Comments on Exercise 2.8: As a hint, here is a proof of a relevant algebraic fact. You will want to incorporate this into your proof. If we have two strings of numbers $a_1 \geq a_2 \geq \cdots \geq a_n$ and $b_1 \geq b_2 \geq \ldots b_n$ then among all sums of products $a_i b_j$, where a is paired with a different b, we find the largest sum is $a_1b_1 + a_2b_2 + \cdots + a_nb_n$. You can prove this by induction. For the base case lets show that if $a_1 \geq a_2$ and $b_1 \geq b_2$, that $a_1b_1 + a_2b_2 \geq a_1b_2 + a_2b_1$. To do this notice that the difference is nonegative:

$$\begin{aligned} a_1b_1 + a_2b_2 - a_1b_2 - a_2b_1 &= a_1b_1 - a_1b_2 + a_2b_2 - a_2b_1 \\ &= a_1(b_1 - b_2) - a_2(b_1 - b_2) \\ &= (a_1 - a_2)(b_1 - b_2) \geq 0, \end{aligned} \tag{1}$$

as both factors are ≥ 0.
Now for the inductive step assume the claim is true for sums of $n-1$ terms and show it is true for sums of n terms. Consider the random sum of a_ib_j terms. If it contains a_1b_1, then it remains only to prove that the remaining terms a_ib_j sum up to less than $a_2b_2 + a_3b_3 + \cdots + a_nb_n$. As this sum has $n-1$ terms, this follows by assumption.
Now assume that a_1b_1 is not in our sum. Then there are some terms a_1b_j and a_kb_1 in it. Then

$$(a_1b_1 + a_kb_j) - (a_1b_j + a_kb_1) = (a_1 - a_k)(b_1 - b_j) \geq 0$$

so we may increase the sum by replacing $a_1b_j + a_kb_1$ with $a_1b_1 + a_kb_j$. Our new sum contains a_1b_1 already, so we have reduced the problem to the previous case.

Comments on Problem 2.1: So far our one item auctions have been "one-shot": we solicit bids and then decide on a winner and a price. The seller in an online auction doesn't have the ability to see everyone's bid first before selling. A deterministic online auction sets prices for bidder i by a function (determined ahead of time) $p_i = f_i(b_1, b_2, \ldots, b_{i-1})$ of the bids before player i. For example the seller could set $p_1 = 100$ and $p_i = b_{i-1}$ for $i > 1$. You can assume that the bidders all know the auctioneer's algorithm (i.e. his functions f_i, for each i, for setting prices) before the auction.

(part b): In part (b) the order bidders arrive is arbitrary. This means that they can work together (i.e. collude). Remember that the bidders know f_i ahead of time. This means that the bidders know what will be the price for the ith bidder based on the first $i-1$ bids. The question asks you to prove that if bidders can arrive in any order the seller cannot find an algorithm that will guarantee him/her getting the max valuation of the bidders.

(part c): In part (c) the order of bidders arrive according to a uniform distribution on the numbers 1,2,. . . ,n. Here is an algorithm that you should show will guarantee the seller gets $c = \frac{1}{4}$ the max valuation no matter how many bidders there are (i.e. independent of n). The seller does not sell to the first n/2 applicants no matter what their value is. She then sets p equal to the largest of the n/2 bids and sells to the next bidder who has value at least p.

This algorithm will definitely return the most valuable element if second most valuable element is in the first n/2 elements and the most valuable element is in the last n/2 elements, since then we will set the threshold p to the second largest value and the best applicant is still available. With what probability does this occur?